



Dillon RL1100 & CP2000 Quick Start

Prepared By:

Nicholas J. Gross, P.E. (Founder)
Gross Engineering LLC
215 Barrington Ridge Lane
Wentzville, Missouri

Revision: 2.0

Date: 2026-6-1

Confidential – Proprietary Information

© Gross Engineering LLC. All rights reserved.
This document contains confidential and proprietary information.
Do not distribute, reproduce, or share without written permission.

IMPORTANT SAFETY INFORMATION (READ FIRST)

General Safety

This system is a powerful mechanical device capable of sudden, forceful movement. Automation enhances speed and consistency but also magnifies the consequences of errors; it does not inherently make reloading safer.

User Responsibility

This system is intended only for responsible adult users who:

- Maintain focus while the machine is operating.
 - Stop the system immediately if anything appears or sounds incorrect.
 - Exercise sound judgment during installation, adjustment, and operation.
- Do not operate this system if you are distracted, impaired, fatigued, or uncertain about the machine's function.

Operating Environment

- Never allow children or untrained individuals to operate the system or be near it while it is powered.
- Always assume the machine is capable of movement when power is applied.
- If unsure about any step, stop and reassess before proceeding.

Warnings

WARNING: Unexpected motion, pinch points, rotating components, and stored energy can cause severe injury or damage. Keep hands, tools, clothing, and loose items clear of all moving components at all times.

WARNING: Never operate the system unattended. Never bypass guards, safety devices, or emergency stop functions.

LIABILITY DISCLAIMER

This product is manufactured by Gross Engineering LLC (referred to as “the manufacturer”).

Reloading ammunition and operating powered machinery are inherently hazardous activities. By installing and using this system, you acknowledge and agree that you are solely responsible for its safe installation, setup, adjustment, and operation, and you assume all associated risks.

The system is an aid and does not substitute for judgment, attention, or responsibility.

The manufacturer is not liable for injury, death, or property damage resulting from: misuse, improper installation, unsafe operation, bypassing safety features, ignoring warnings, user modifications, third-party components, or unsupported configurations.

Quick Start Guide

This guide covers the assembly and initial setup of the Dillon 1050 AutoDrive by Gross Engineering.

It is intended to be used alongside the full technical manual and provides a fast, simplified path to get your system up and running.

Included Components

- (1) Aluminum base plate (12" × 8")
- (1) Worm gearbox (1:20)
- (2) L-shaped mounting brackets
- (1) 1100 Replacement Crankshaft

Hardware

- (4) M6 × 16 flange bolts (black oxide)
- (4) M6 × 20 flange bolts (black oxide)
- (4) M6 × 30 socket head cap screws (Black Oxide)
- (4) M6 x 40 socket head cap screw (Blue)
- (8) M6 nyloc nuts
- (4) M6 washers
- (1) M8 Bolt

Shafts & Keys

- (2) Small keys
- (2) Large key
- (1) M18 keyed gearbox shaft
- (1) M18 to M18 clamping shaft

Sensors & Brackets

- (1) Proximity bracket (sheet metal)
- (2) M12 inductive proximity sensors
- (1) 1100 flag
- (1) Laser / IR bullet sensor + bracket

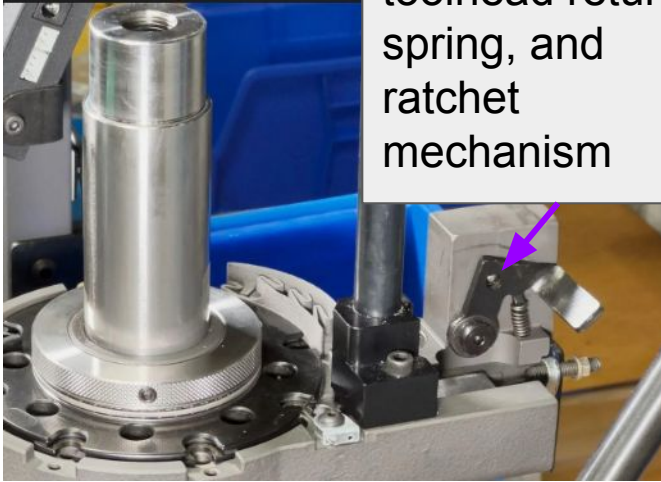
Electronics

- (1) NEMA 34 stepper motor (keyway included, packaged in red bag)
- (1) Touchscreen HMI
- (1) 9-pin D-sub cable
- (1) Control box
- (1) AC power cable

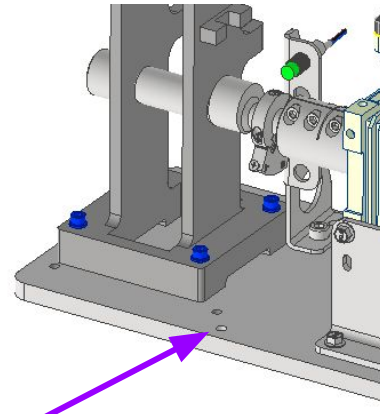
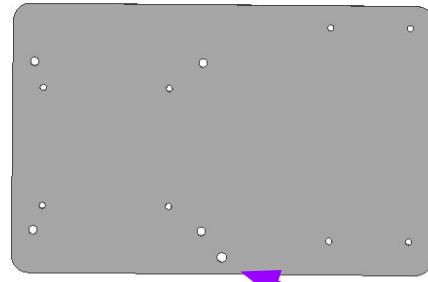


1.

Remove toolhead, toolhead return spring, and ratchet mechanism

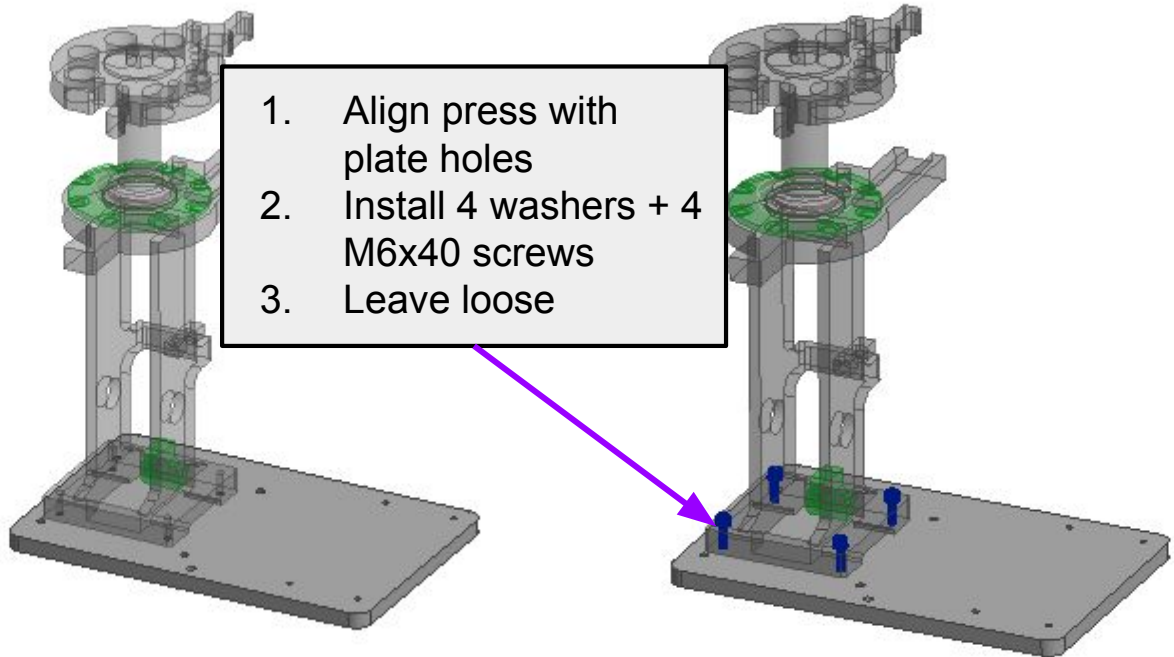


2.



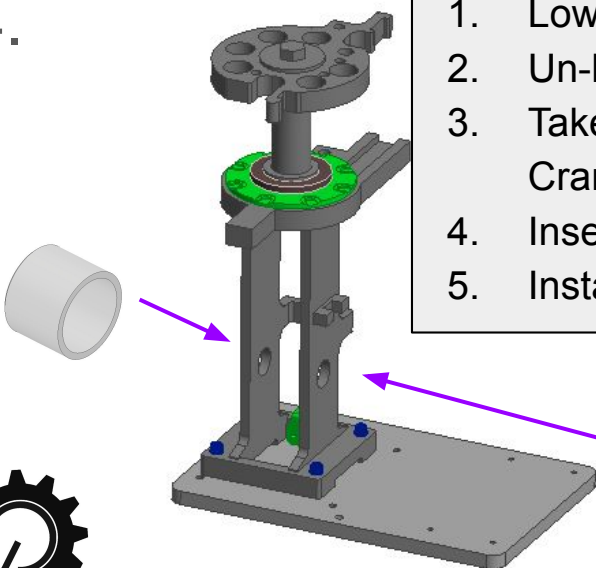
Orient plate so non-thread hold is in front as shown

3.

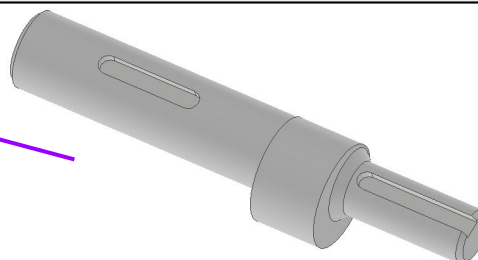


1. Align press with plate holes
2. Install 4 washers + 4 M6x40 screws
3. Leave loose

4.



1. Lower piston to the bottom
2. Un-bolt and remove the OEM crankshaft.
3. Take the oem Key and place it on the New Crankshaft
4. Insert the Crankshaft
5. Install Bearing Cap.

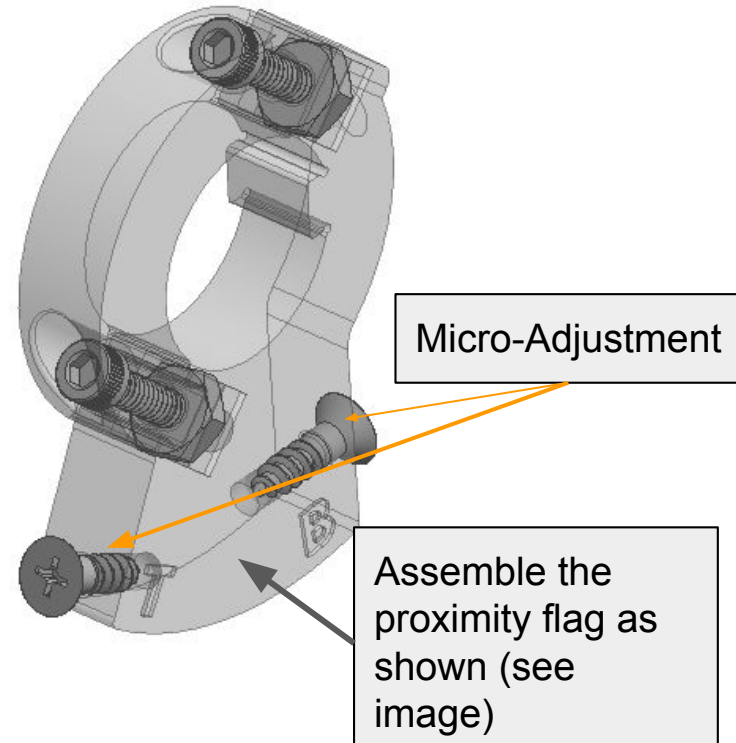


5.

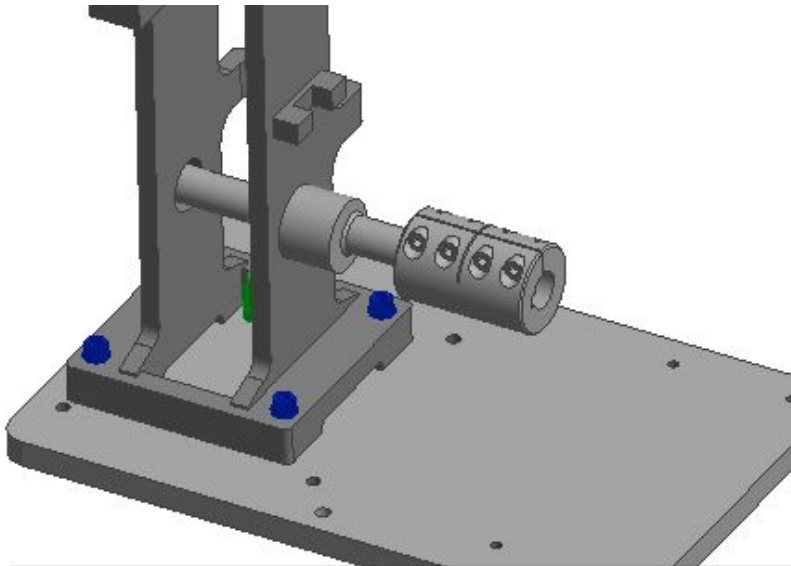


1. Install M12 sensors into bracket
2. Use supplied M12 nuts to secure

6.

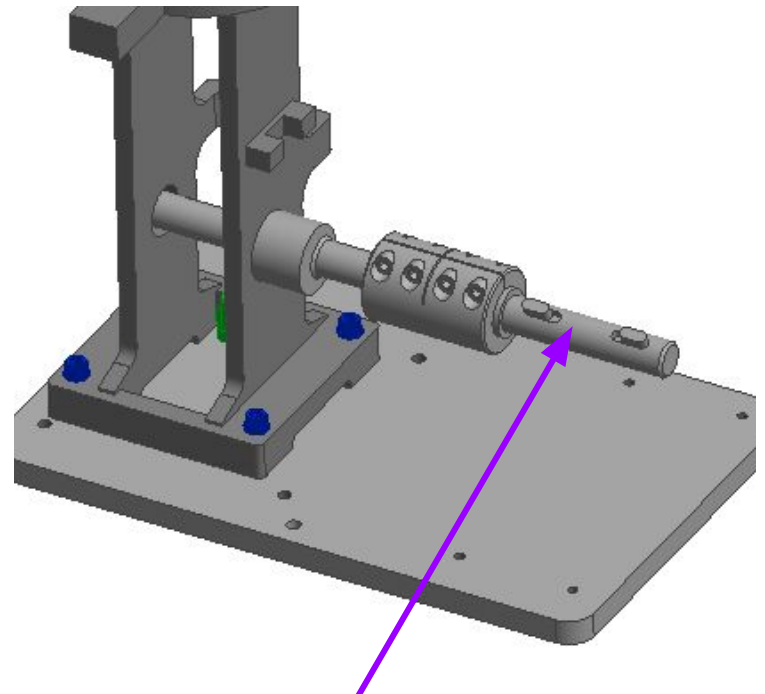


7.



1. Install 18 x 18 shaft coupler,
2. Install 6mm key stock
3. Tighten press side of the coupler

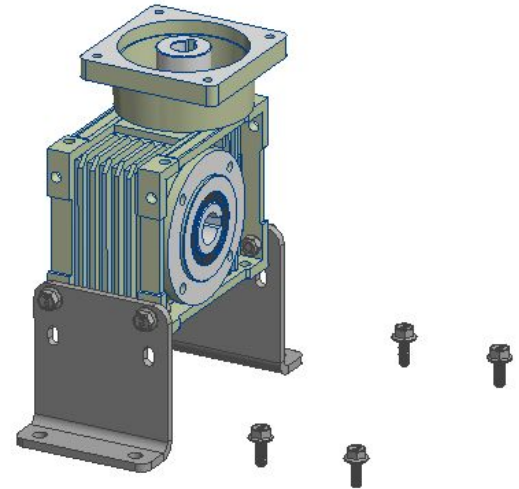
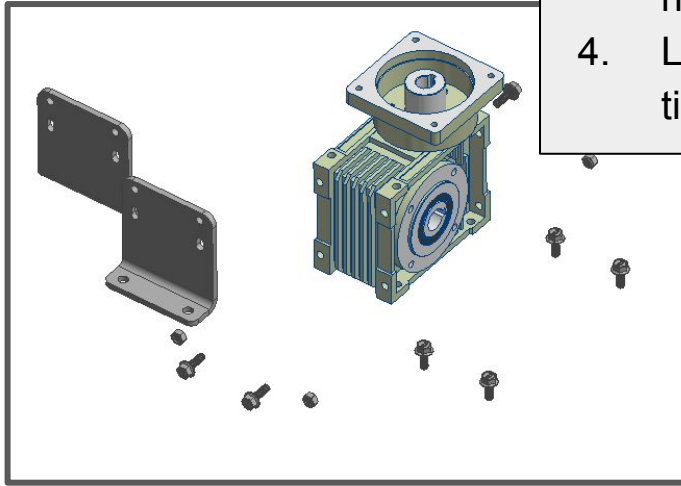
8.



1. Install 18 gearbox shaft with the 6m key stock into the coupler
2. Tighten
3. Place both small Key stock

9.

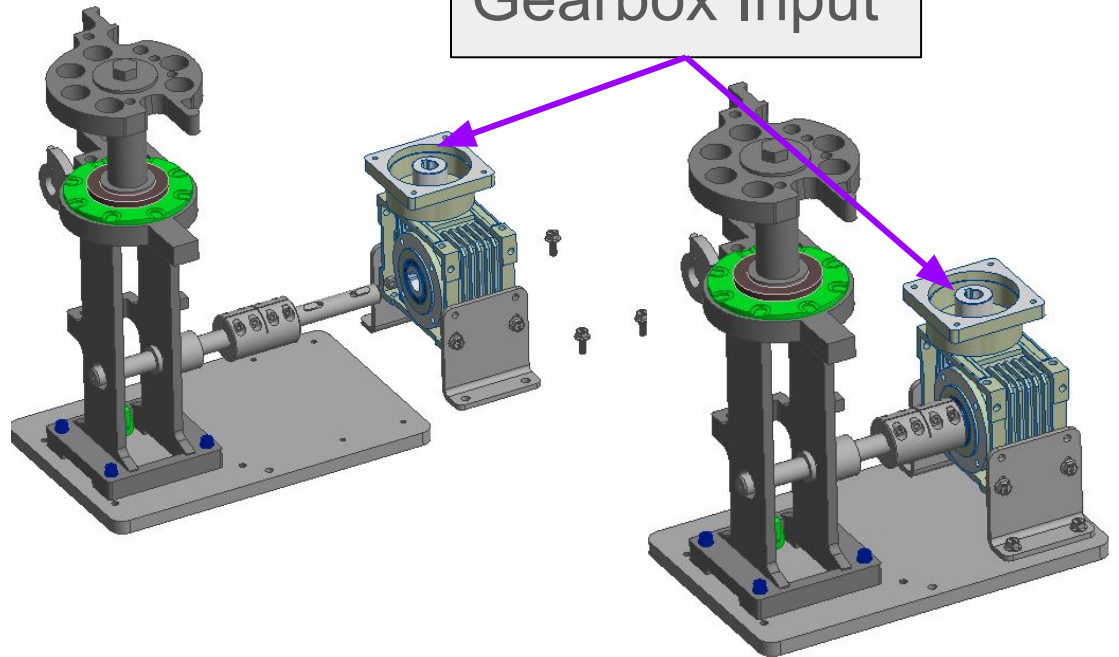
1. Align both L-shaped brackets (see image)
2. Install four M6 × 20 flange bolts in the shown holes
3. Secure with four nyloc nuts
4. Leave loose (do not tighten)



10.

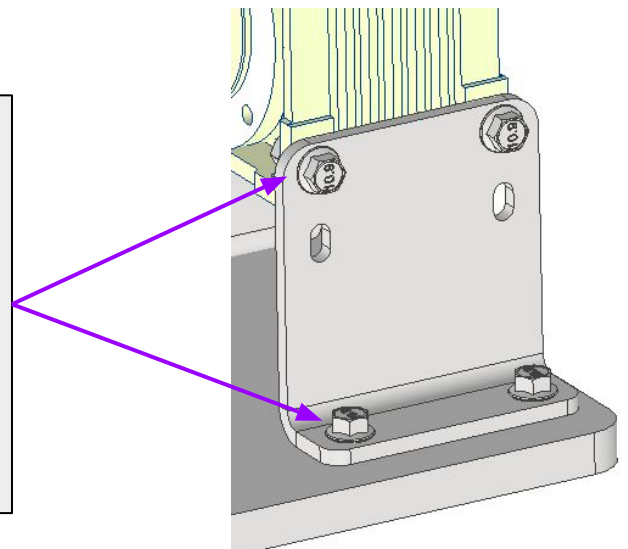
1. Align keyway with shaft keys
2. Rotate **input shaft only** (do not move gearbox)
3. Slide gearbox onto shaft
4. Align L-shaped bracket mounting holes with baseplate
5. Loosen gearbox mounts if needed to align
6. Leave loose (do not tighten)

Gearbox Input

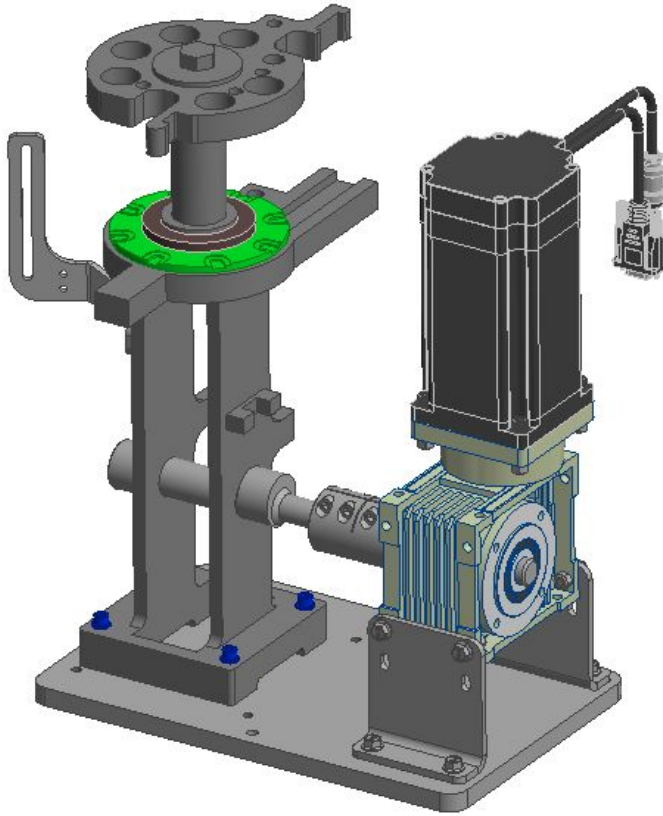


11.

1. Install M6 × 16 flange bolts through the L-brackets into the baseplate
2. Loosen other fasteners as needed for alignment
3. The gearbox will self-align to the press
4. Ensure all gearbox and mount bolts remain loose (do not tighten)

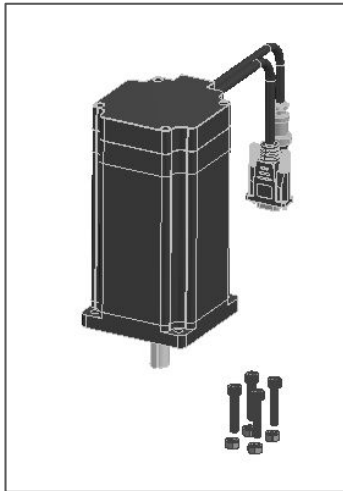


12.

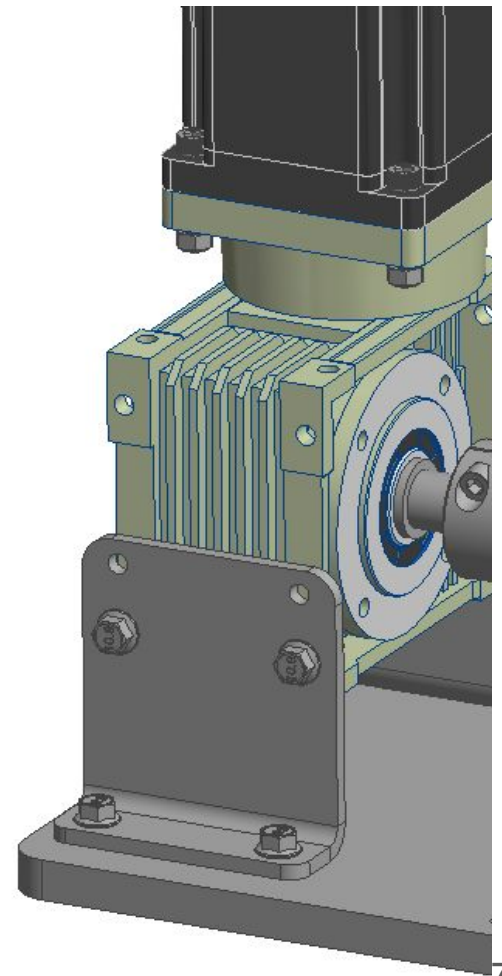
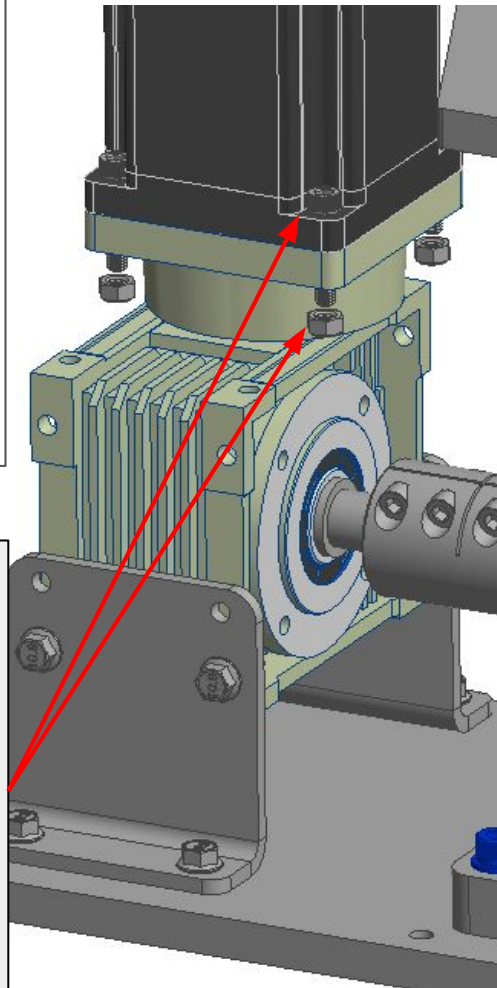


1. Start with all components aligned (see image)
2. Tighten the four M6 bolts holding the press down
3. Tighten the four clamping collar bolts
4. Tighten the L-brackets to the baseplate
5. Tighten the gearbox to the L-brackets
6. If resistance occurs, loosen bolts and realign

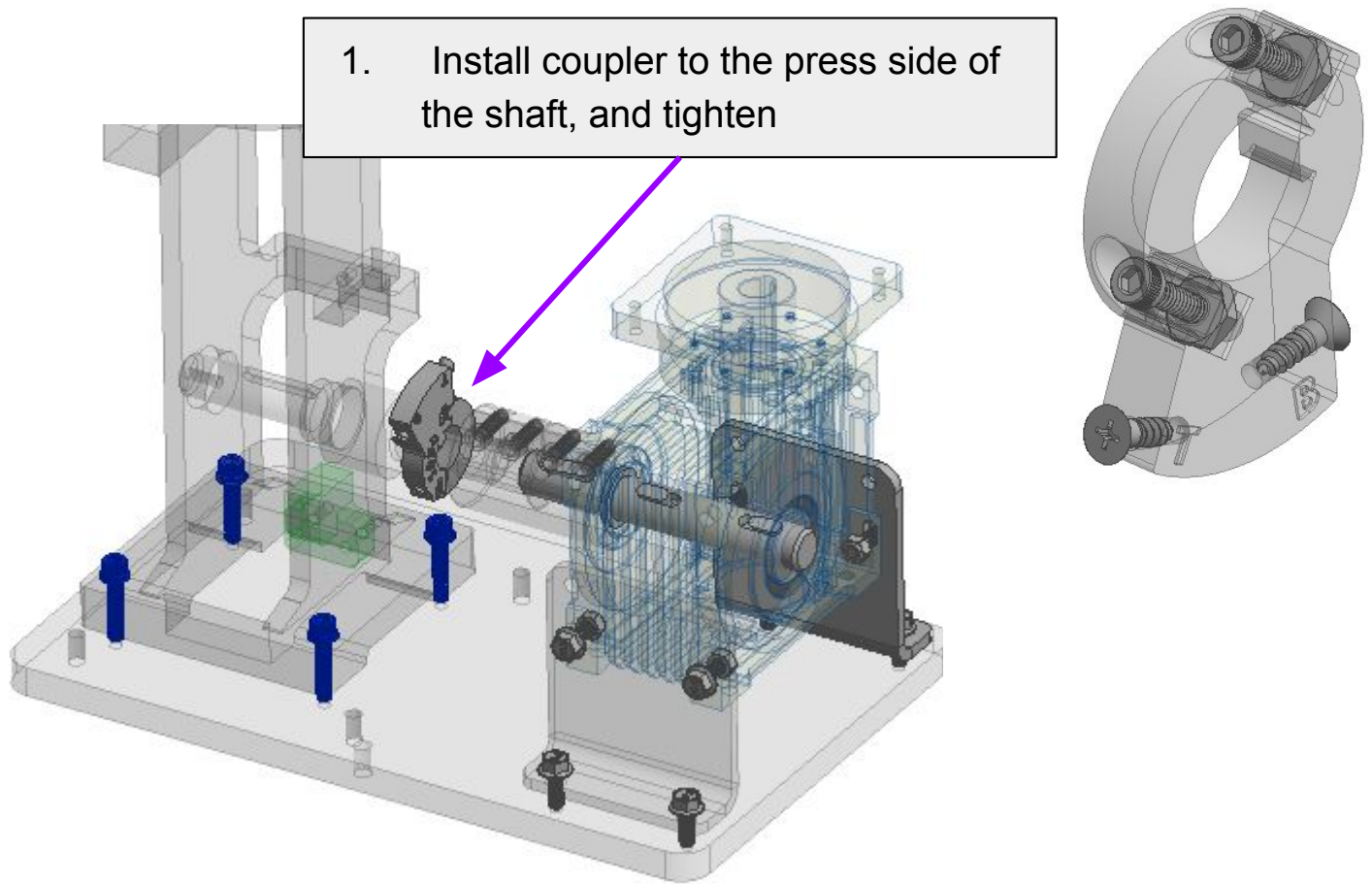
13.



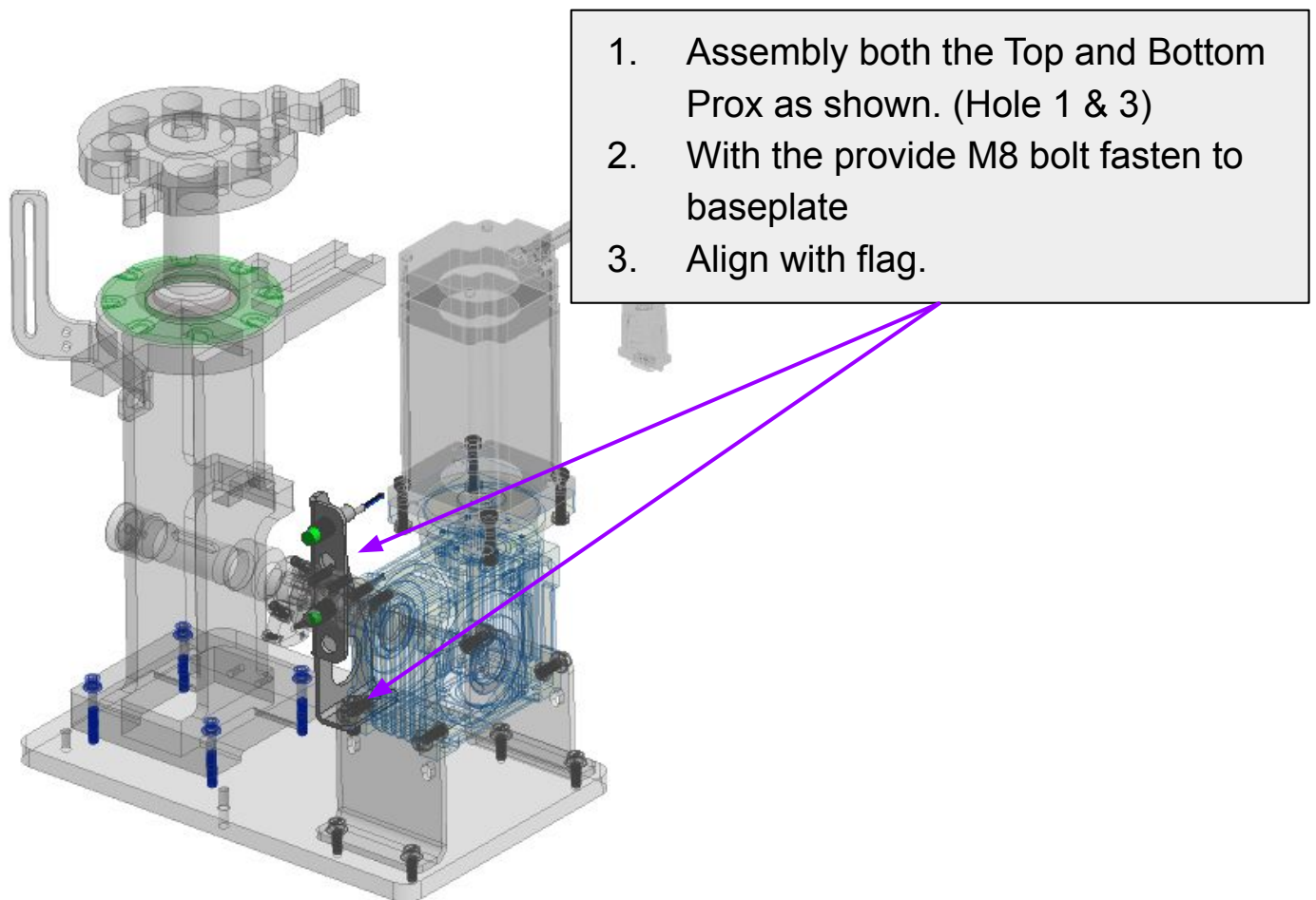
1. Align motor shaft key with gearbox keyway
2. Insert motor into gearbox
3. Install four M6 × 30 bolts
4. Secure with four nyloc nuts
5. Tighten until snug



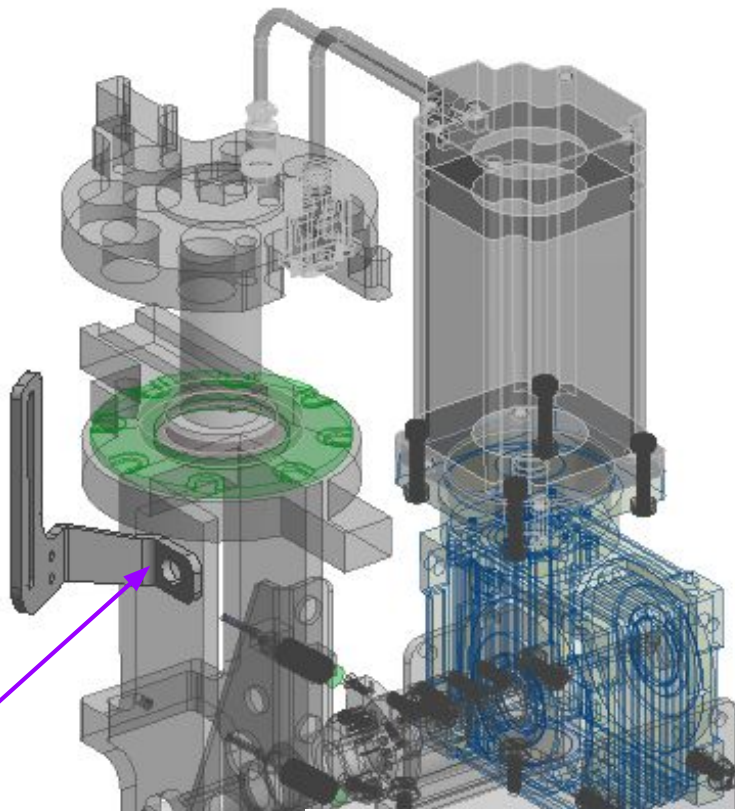
14.



15.



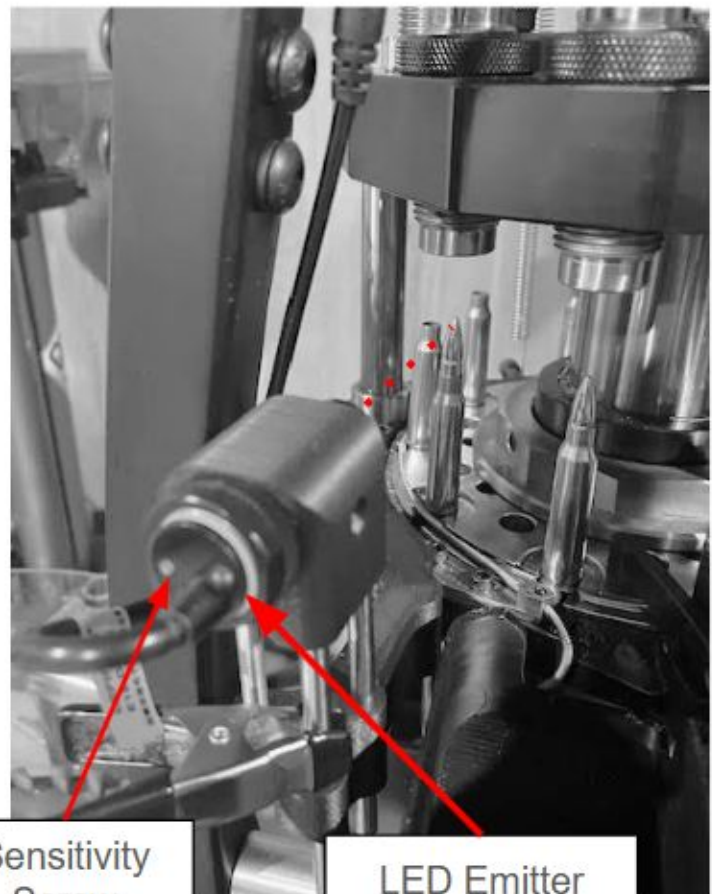
16.



1. Mount the sensor bracket to the base frame
2. Tighten and secure
3. Lower the sensor until the indicator light turns on

Bullet Sensor Setup

- Mount sensor at Station 7 (before bullet seating)
- Align sensor horizontal ("like sunglasses")
- Place case with bullet
- Turn screw **counterclockwise** until LED turns off
- Turn **clockwise** until LED just turns on
- Adjust height so LED turns off when:
 - No bullet
 - Bullet is misaligned
- Test good vs bad cases
- Confirm consistent detection
- Disable sensor in Advanced if not in use
- Take time to tune to avoid false faults

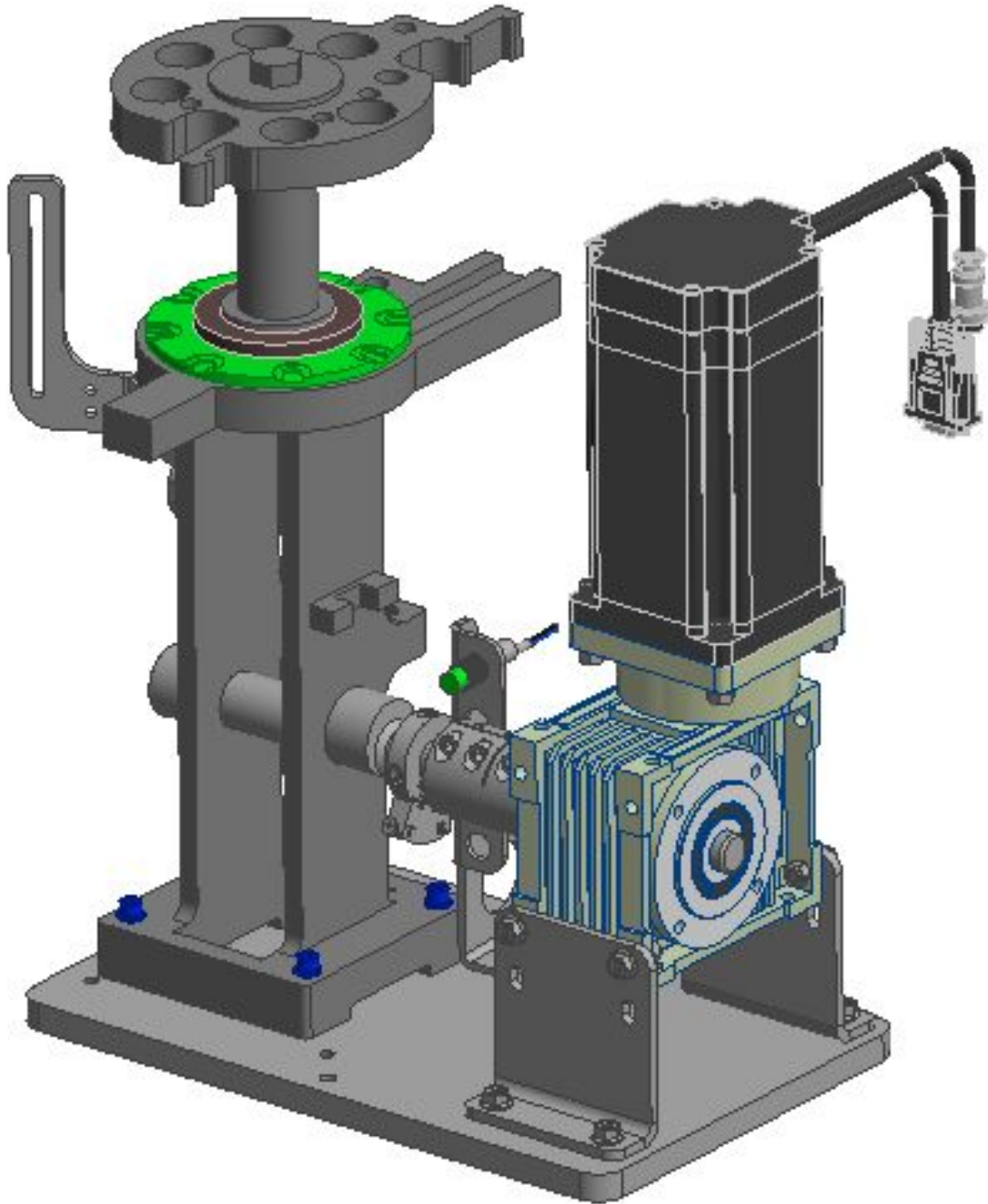


Sensitivity
Screw

LED Emitter

17.

1. Verify system matches the image
2. Ensure all bolts and fasteners are fully tightened
3. Confirm top and bottom hard stops are removed
4. Store removed parts with unused components

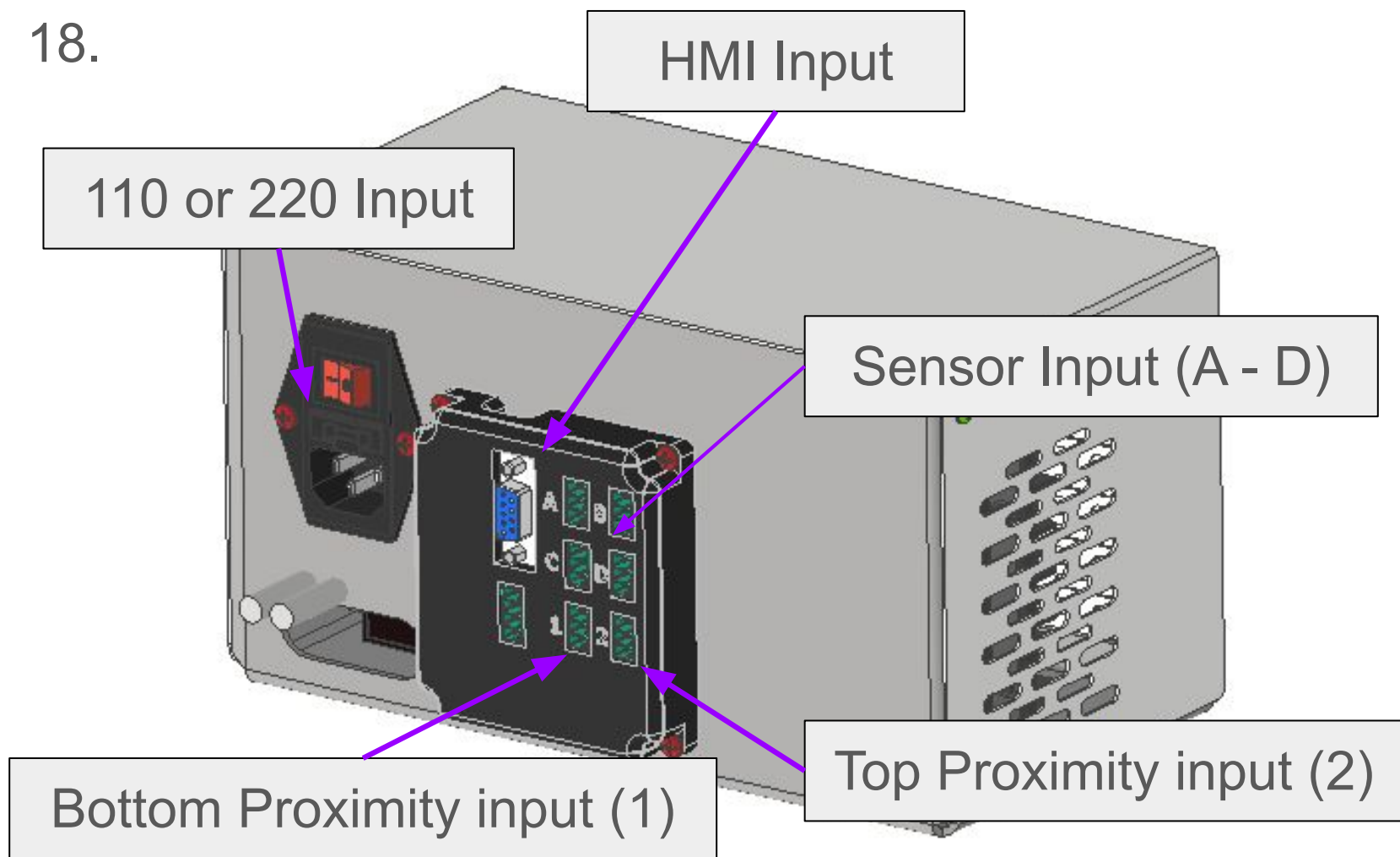


Confidential – Proprietary Information

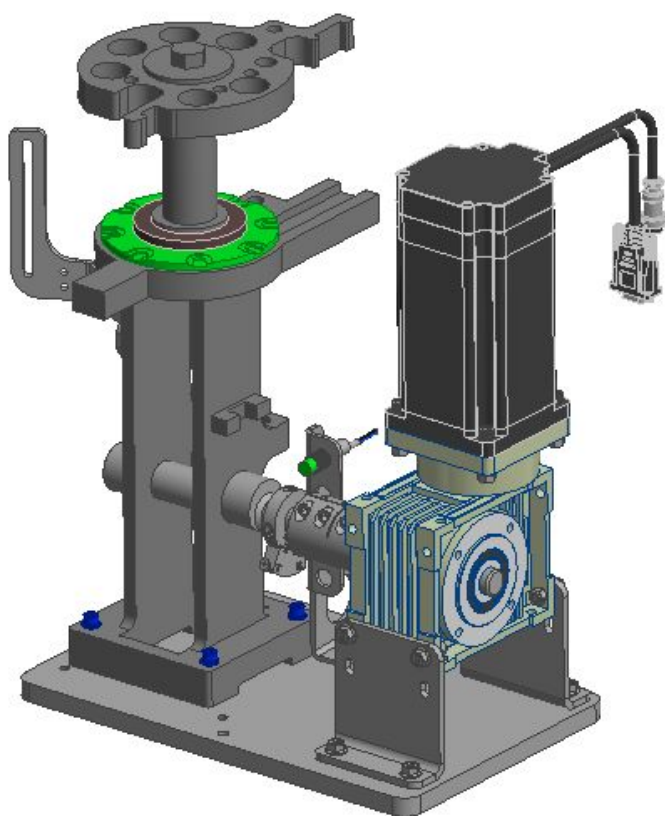
© Gross Engineering LLC. All rights reserved.

This document contains confidential and proprietary information.
Do not distribute, reproduce, or share without written permission.

18.



19.



1. Connect 15-pin encoder cable from stepper motor to control box
2. Connect 4-pin power cable from motor to control box
3. Connect bottom proximity sensor to Port 1
4. Connect top proximity sensor to Port 2
5. Connect 9-pin D-sub cable from HMI to control box

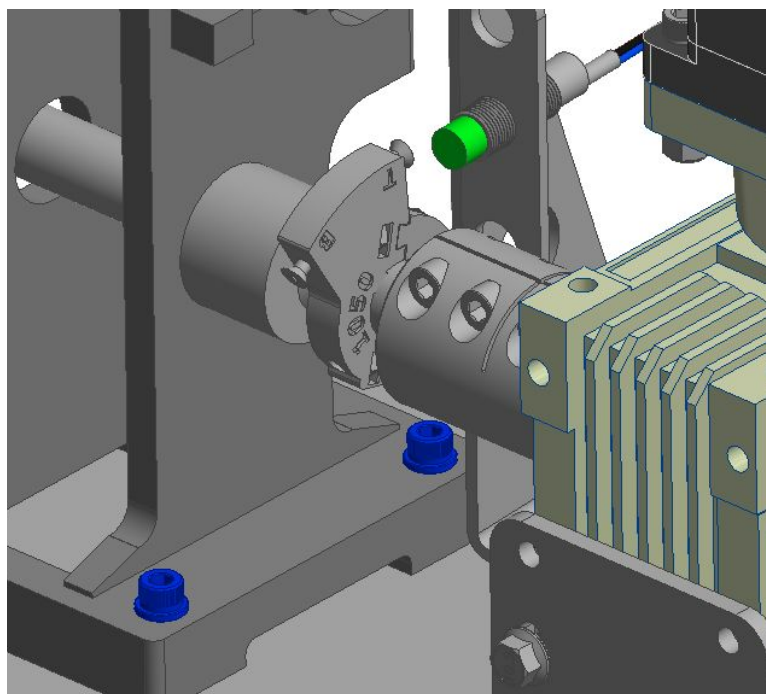
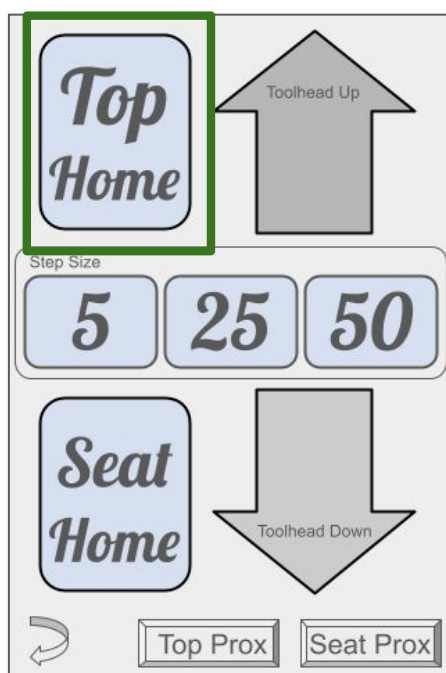
Power On Check

1. Turn on the machine
2. Verify system powers up correctly

Top Proximity Sensor Setup



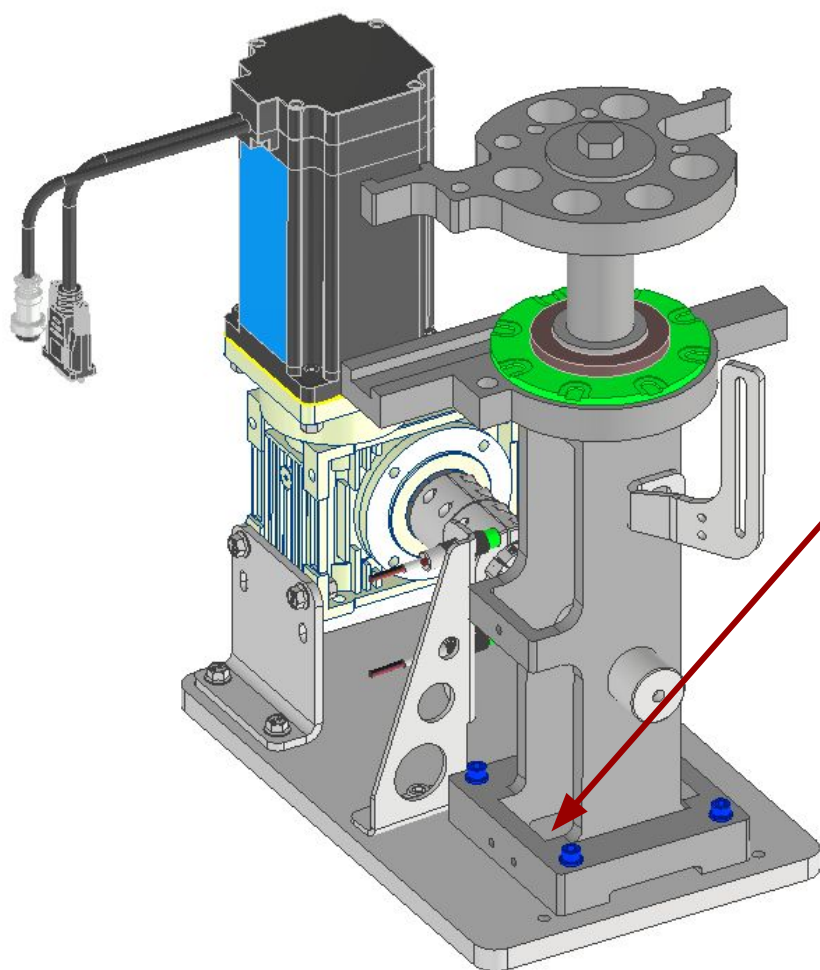
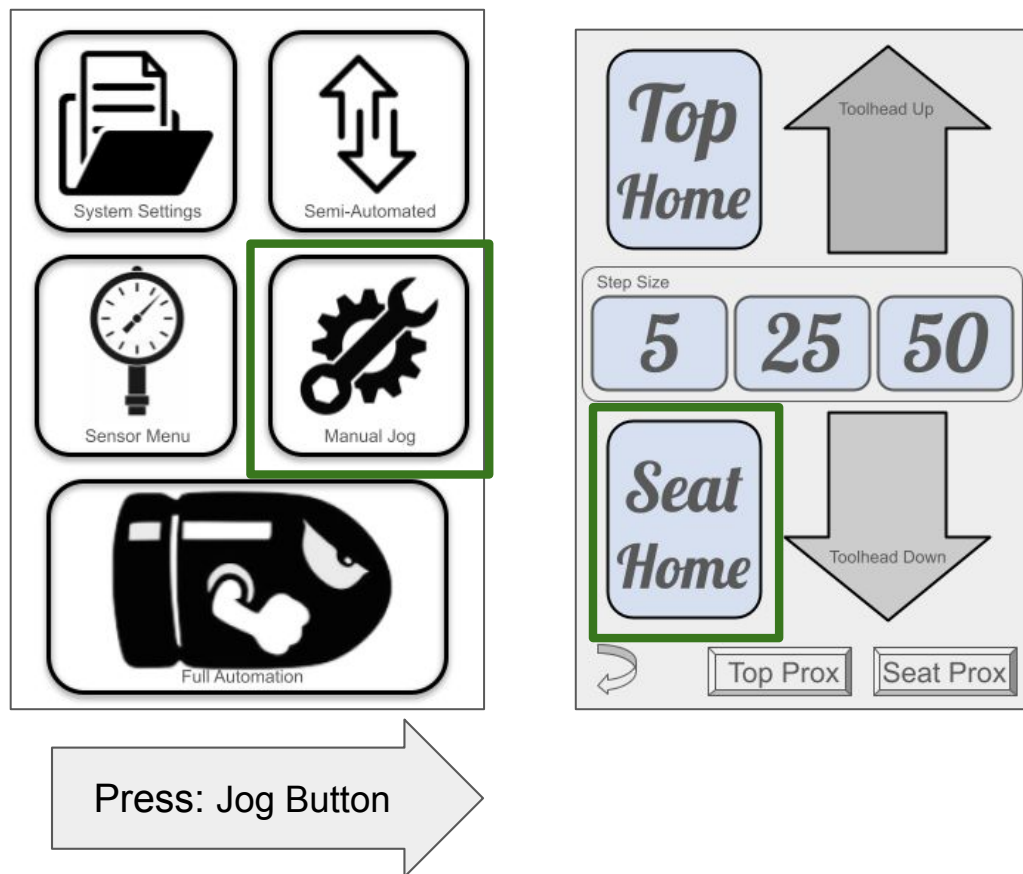
Press: Manual Jog
(file icon)



Set Top Proximity Sensor

1. On the Manual Jog screen, move the press **up**
2. Select step size (50, 10, or 5)
3. Use **50** until close to the top and switch to 10 or 5
4. Watch the shell plate and ensure a shell plate fully indexes.
5. Move the top proximity toward the flag until the indicator light turns off.
6. Press **50** and jog **down** until clear of the sensor
7. Press **TOP home**
8. The press will move up to the sensor
9. Repeat until indexing is clean and consistent

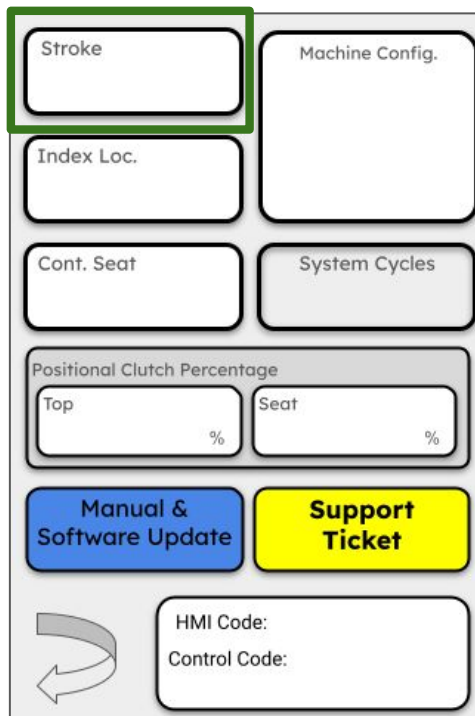
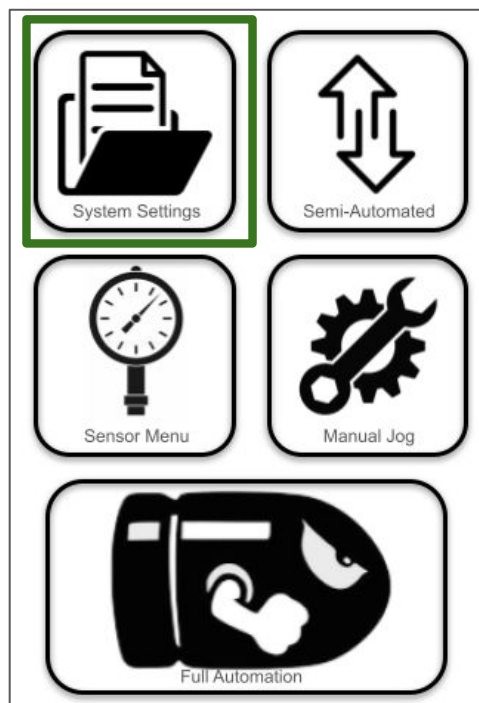
Bottom Proximity Sensor Setup



Set Bottom Proximity Sensor

1. On the Manual Jog screen, move the press **down**
2. Start with step size **50**
3. Switch to **10** and **5** as you get close
4. Stop when the gap is **.0625 to .125**
(Measure from the lower hard stop)
5. Move the bottom sensor inward
6. Stop when the indicator light turns **off**
7. For fine adjustment, turn the sensor body as needed
8. Jog **up** until clear of the sensor
9. Press **Seat Home**,
10. The press will move down to the sensor
11. Repeat until consistent

Stroke Setup



Press: Setting (file icon)

Press
Stroke
Button

Learn Stroke

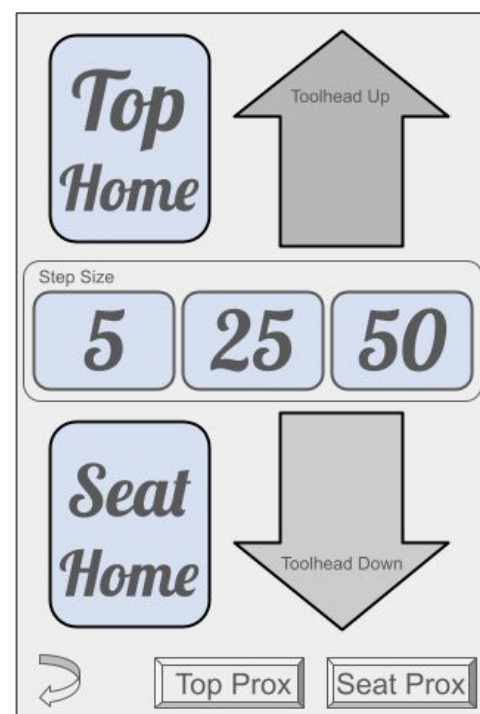
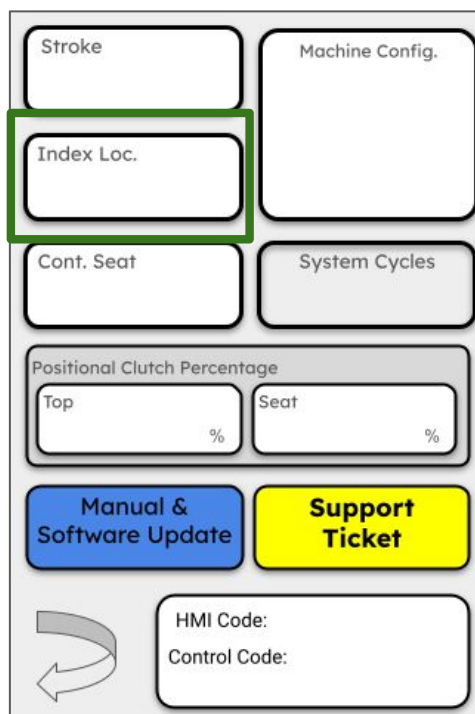
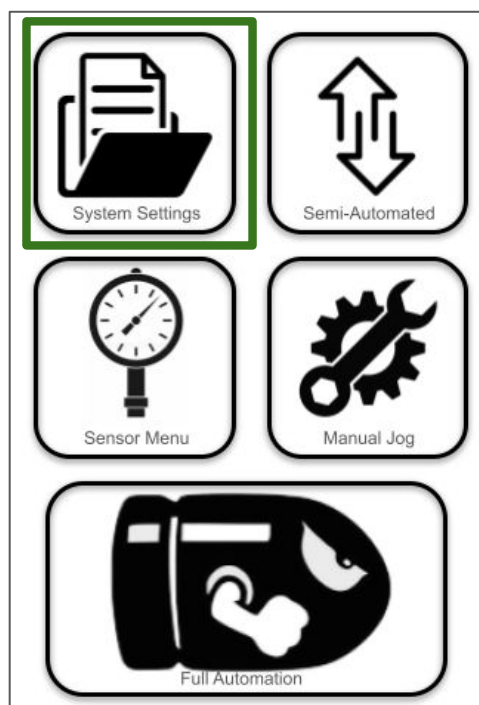
1. Ensure press is empty (no cases)
2. From the main screen, go to **Settings** → **Stroke Setup**
3. Select **Press Text Box**
4. The press will move to the top position
5. Then move from top to bottom to learn the stroke
6. Then move from bottom to top to confirm
7. Stroke is now saved in the system
8. Relearn stroke only if sensors are moved

Confidential – Proprietary Information

© Gross Engineering LLC. All rights reserved.

This document contains confidential and proprietary information.
Do not distribute, reproduce, or share without written permission.

Index Setup



Press: Setting (file icon)

Press: Index (file icon)

Press: Home

Set Index Position

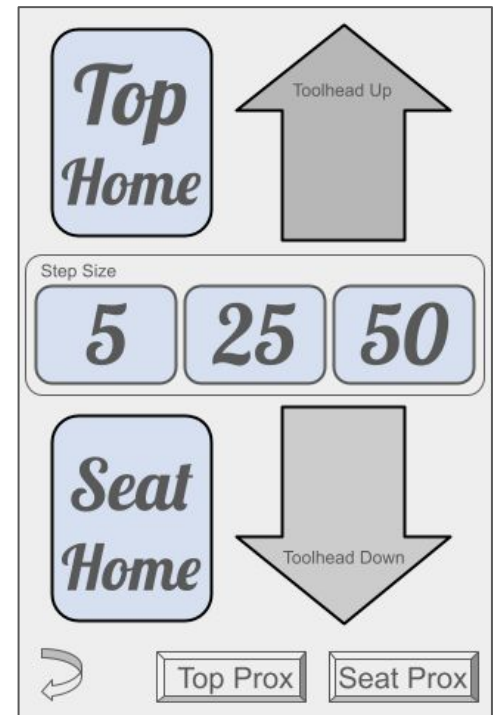
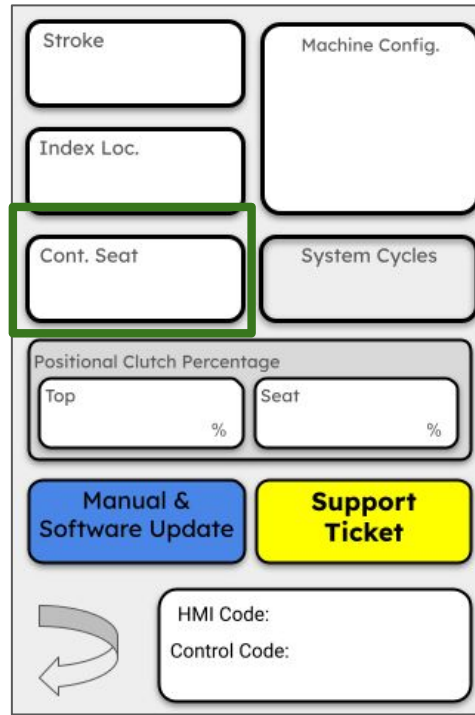
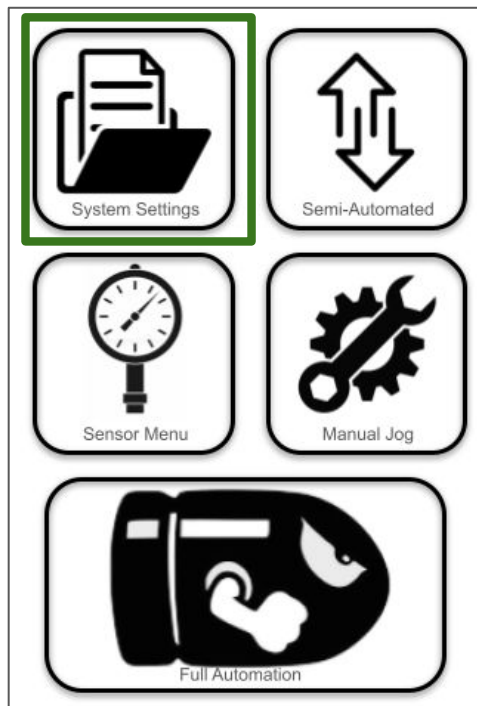
1. Go to **Settings** → **Index Dialog**
2. Press **Press Dialog Box**
3. System will open the Manual Jog screen
4. Use **50** → **10** → **5** and arrows to move the press
5. Jog up or down to find the index position
6. Listen for the audible click
7. Position does not need to be exact
8. Set it where you want indexing to occur
9. Press either **Seat Home**
10. The press will cycle and return to the set position
11. Repeat until position is correct

Confidential – Proprietary Information

© Gross Engineering LLC. All rights reserved.

This document contains confidential and proprietary information.
Do not distribute, reproduce, or share without written permission.

Control Seat Setup



Press: Setting (file icon)

Press: Primer (file icon)

Press: Home

Set Primer / Controlled Seat

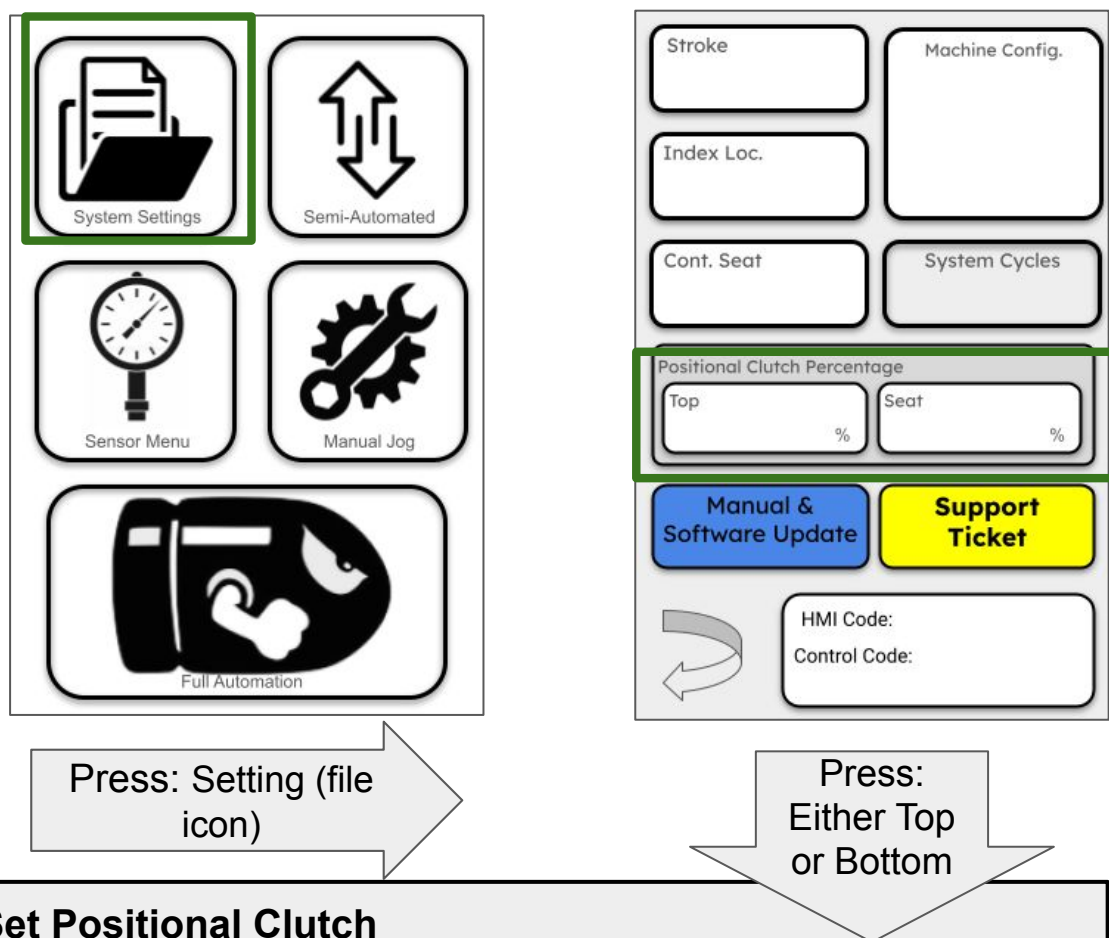
1. Go to **Settings** → **Primer / Seat Set**
2. Press **Start**
3. System will open the Manual Jog screen
4. Use **50** → **10** → **5** and arrows to move the press
5. Jog to the point where primer insertion should begin
6. Position can be set slightly **before or after** contact
7. Choose what works best for your setup
8. Press **Home**
9. The press will cycle and return to the set position
10. Repeat until position is correct

Confidential – Proprietary Information

© Gross Engineering LLC. All rights reserved.

This document contains confidential and proprietary information.
Do not distribute, reproduce, or share without written permission.

Clutch Setup



Set Positional Clutch

1. Go to **Settings** → **Clutch Setup**
2. Select **Top** or **Bottom** value
 - Enter a whole number using the keypad
3. Start with **10%** and adjust as needed
 - Lower = tighter (less movement)
 - Higher = more forgiving (more movement)
4. Press **Enter**,
 - Wait for "Recording" to finish

Key Notes

- Clutch value = stroke X % of allowed motion
- Does **not** add or reduce torque
- Allows the machine to struggle longer before stopping
- Machine will **not** drive past proximity sensors (hard stop)
- Repeat for both Top and Bottom
- If all values show **0** (including stroke):
 - HMI is not communicating
 - Power cycle the machine

Confidential – Proprietary Information

© Gross Engineering LLC. All rights reserved.

This document contains confidential and proprietary information.
Do not distribute, reproduce, or share without written permission.

Sensor Setup

Sensor Configuration

1. From the **Main Menu**, press **Sensors**.
2. Select **Port A, B, C, or D** to configure.
3. Press **Assign Sensor** and select the desired sensor from the list.
4. Press the **Back Arrow** to save the sensor assignment.

Sensor Settings

- **Active** = Sensor is enabled and available for use.
- **Deactivated** = Sensor is assigned but disabled.
- Select whether the sensor is evaluated with the **Tool Head Up** or **Tool Head Down**.

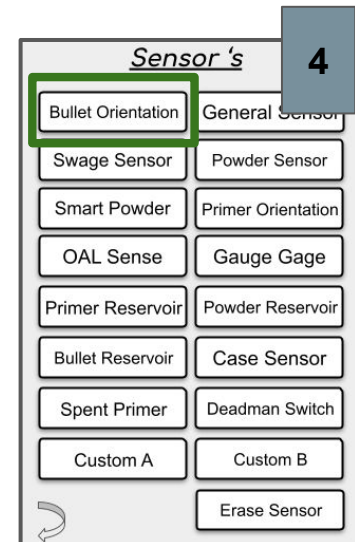
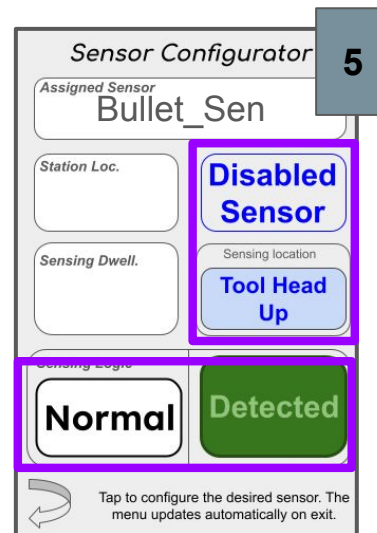
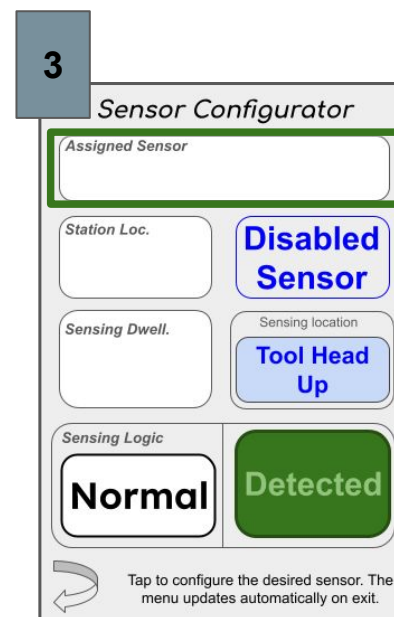
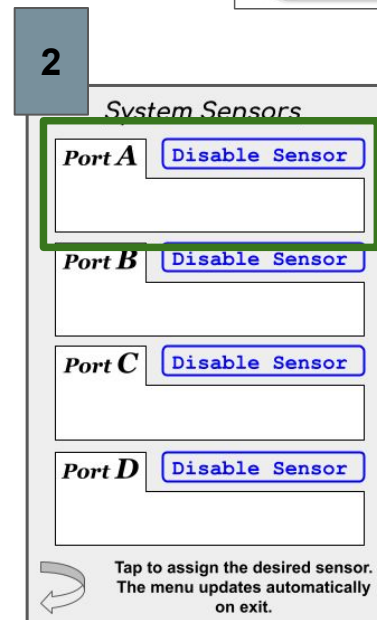
Sensor Logic

- Verify the sensor is indicating a **Green PASS** condition during normal operation.
- **Green PASS** = The sensor is in the expected state and will not generate an error.
- **Blue FAIL** = The sensor is not in the expected state and will generate an error if the machine evaluates that condition.
- If the sensor is displaying **Blue FAIL** when it should be passing, switch between **Normal** and **Inverse** logic and verify operation again.

Saving

- Press the **Back Arrow** to save all settings.
- The assigned sensor name will appear next to the selected port.
- Repeat for any additional sensors.
- All settings are retained after power cycling.

Example: Assign a **Bullet Present** sensor, set it to **Active**, and configure it to be evaluated with the **Tool Head Up**. Adjust the **Normal/Inverse** logic until the sensor displays a **Green PASS** when a bullet is present. The sensor will then be available for use when configuring a load.



Confidential – Proprietary Information

© Gross Engineering LLC. All rights reserved.

This document contains confidential and proprietary information.
Do not distribute, reproduce, or share without written permission.

Mark7 Sensor Adapter Setup

Sensor Wiring

This system uses a standard wiring layout for all sensors.

- Same wiring applies to all ports
- Supports terminal blocks and Molex connectors

Terminal Block Wiring

Two connection types:

- **Screw Terminal** – tighten with screwdriver
- **Push Terminal** – insert wire (no tools)

Wiring Layout (All Ports):

- **Position 1** – Black → Signal
- **Position 2** – Blue → Ground (-)
- **Position 3** – Brown → Power (+)

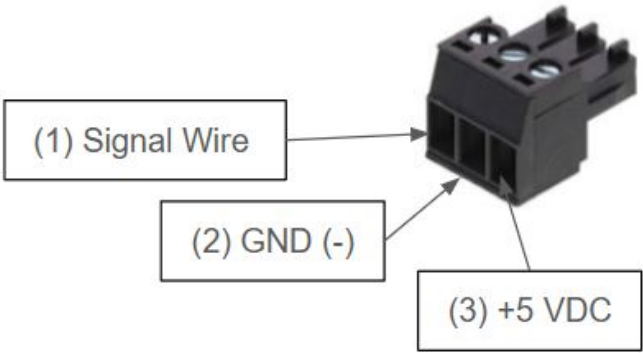
Molex Connector (Mark 7)

Some sensors use Molex MicroFit connectors.

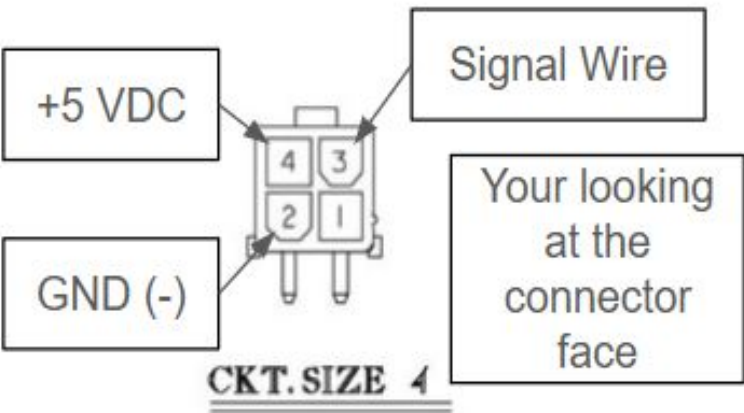
- Do **not** cut or modify the Mark7 sensor wire
- Use adapter or match wiring as shown

Molex to Terminal Mapping:

- **Pin 3** → **Position 1** (Black – Signal)
- **Pin 2** → **Position 2** (Blue – Ground)
- **Pin 4** → **Position 3** (Brown – Power)



5 Sets 3-Pin (3 Pole) 3.81mm Pitch Right Angle Block Connector PCB Mount DIY
Brand: DBParts
4.5 ★★★★★ (12)
\$7.91
Get Fast, Free Shipping with Amazon Prime
FREE Returns
Current Rating: 8 Amps
Item Pitch: 3.81 Millimeters
Gauge: 12.0
Connector Type: Screw
Number Of Contacts: 3
Mounting Type: PCB Mount
Material: Copper
See more



Adapter Cable

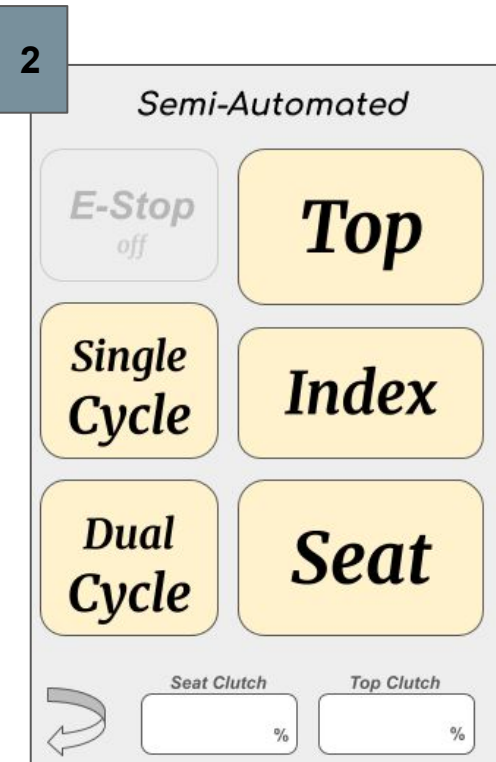
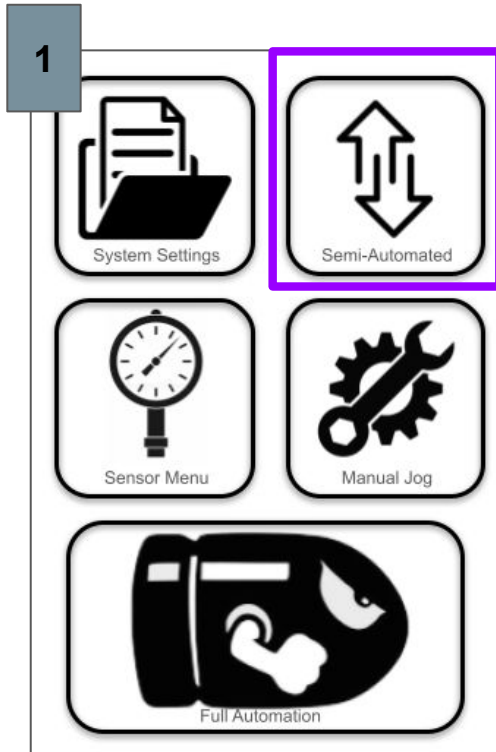
Product features
ATX 4 Pin CPU Power Cable, 4Pin CPU Male to Female Extension Cable for Desktop Computer Power Supply (2 Pack, 20cm/8inch)
Brand: DKARDU
4.4 ★★★★★ 4 ratings
\$6.99 (\$1.75 / count)
Get Fast, Free Shipping with Amazon Prime
FREE Returns
Up to 10% off if you qualify Shop items
Get \$80 off instantly: Pay \$0.00 upon approval for the Amazon Store Card.
Brand: DKARDU
Connector Gender: Male-to-Female

Confidential – Proprietary Information

© Gross Engineering LLC. All rights reserved.

This document contains confidential and proprietary information.
Do not distribute, reproduce, or share without written permission.

Semi- Automated (Load Development Menu)



Semi-Automated Mode

Semi-Automated Mode allows manual control of the press for setup, testing, maintenance, and other fine-tuning operations.

Entering Semi-Automated Mode

1. From the Main Menu, press the **Up/Down Arrow** icon.
2. The press will automatically move the tool head to the **Top** position.
3. This establishes a safe starting position.

Moving the Press

- Press **Top**, **Index**, or **Seat** to move the press to that position.
- Positions may be selected in any order.
- Press **Single Stroke** to perform one complete cycle.
- Press **Dual Stroke** to perform two complete cycles.

The cycle will begin from the current position and return to that position when complete.

Emergency Stop

- Pressing **E-Stop** at any time immediately stops all motion.
- The E-Stop indicator will illuminate.
- To resume operation, twist and release the E-Stop button.

Exiting Semi-Automated Mode

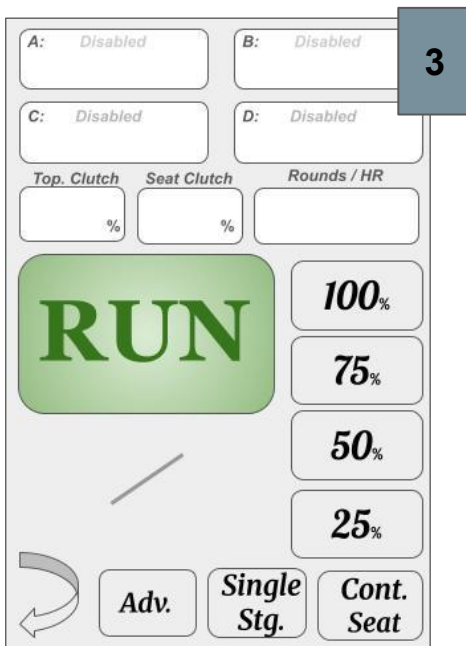
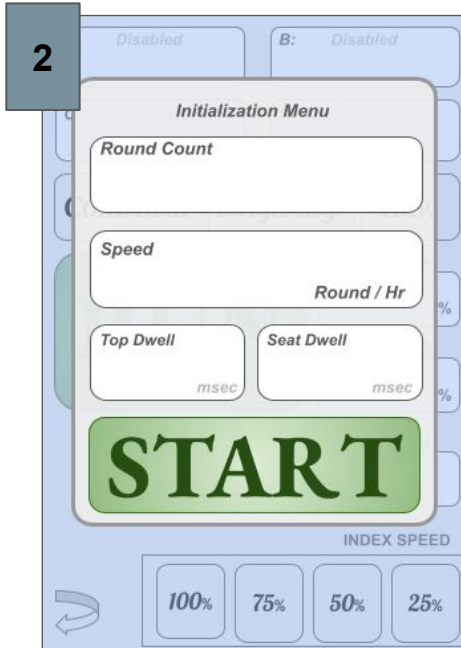
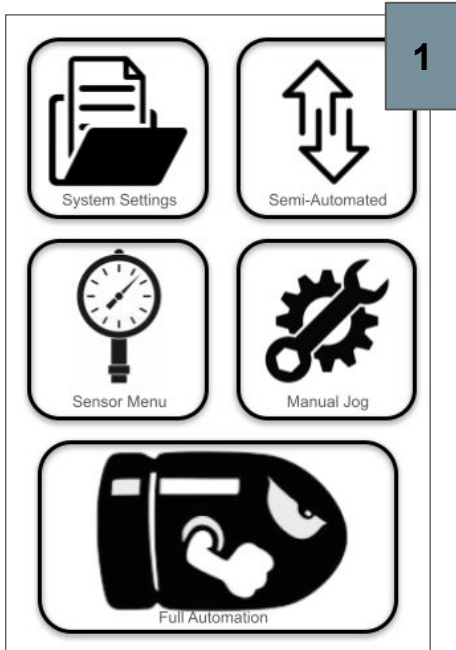
- Press the **Back Arrow** to return to the Main Menu.

Confidential – Proprietary Information

© Gross Engineering LLC. All rights reserved.

This document contains confidential and proprietary information.
Do not distribute, reproduce, or share without written permission.

Automated Run (Standard)



Automatic Mode

Automatic Mode is used for normal production and automated loading.

1. From the Main Menu, press the **Bullet** icon.
2. The tool head will automatically begin homing to the **Top** position.
3. While homing, you may adjust:
 - **Round Count** (default: 10,000)
 - **Speed (RPH)** (default: 1,000)
 - **Top Dwell**
 - **Bottom Dwell**
4. Touch any value to open the number pad and enter a new value.
5. Once homing is complete, press **Start** to enter the Run Screen.

Run Screen

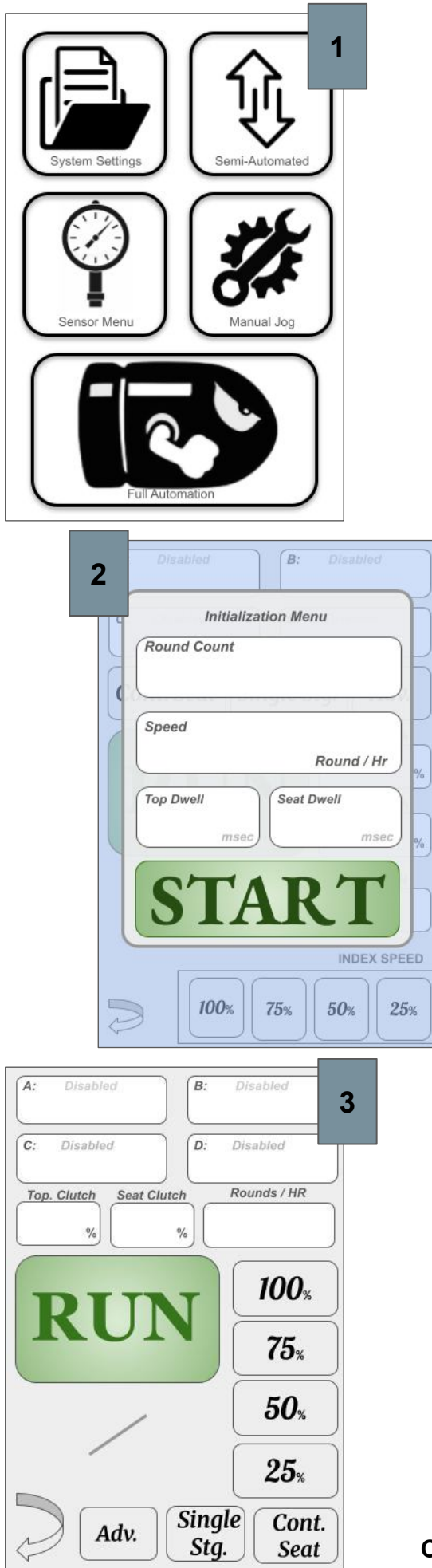
- All assigned and active sensors will be displayed.
- Current machine status and cycle information will be shown.
- Press **Run** to begin automated loading.
- Press **Pause** to stop operation. The machine will finish the current cycle before stopping.
- To exit, press the **Back Arrow** in the lower-left corner. The machine will safely stop and return to the previous screen.

Confidential – Proprietary Information

© Gross Engineering LLC. All rights reserved.

This document contains confidential and proprietary information.
Do not distribute, reproduce, or share without written permission.

Automated Run (Advance features)



Advanced Run Settings

Once on the **Run Screen**, several settings can be adjusted to customize machine operation.

- Sensors assigned to Ports **A, B, C, and D** may be enabled or disabled during a run.
- The system displays real-time **Top Clutch, Bottom Clutch, and Rounds Per Hour (RPH)** values.
- The **Index Speed** button controls how quickly the tool head indexes during the upper portion of the stroke.
 - Available settings: **100%, 75%, 50%, and 25%**
 - Default setting: **100%**
- The **Control Seat** button slows the seating portion of the stroke for additional control when needed.
- **Single Stage** mode performs one complete cycle at a time and is useful for setup, testing, and troubleshooting.
- Advanced features can be enabled or disabled by touching the corresponding button.

Sensor Indicators

When a sensor is enabled and within its monitoring window:

- **Green** indicates the sensor is in the expected state.
- **Blue** indicates the sensor is not in the expected state and may generate an error condition.

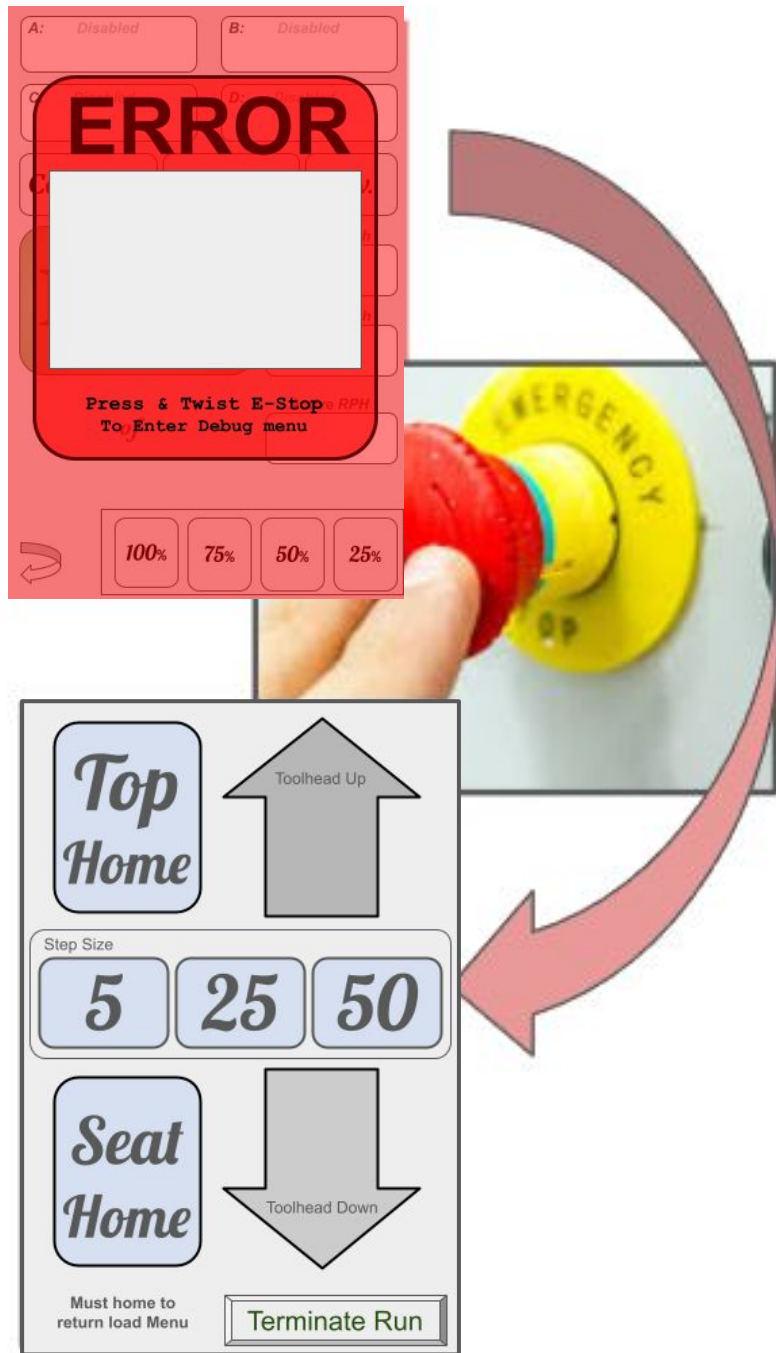
Sensor status is displayed in real time while the machine is operating.

Confidential – Proprietary Information

© Gross Engineering LLC. All rights reserved.

This document contains confidential and proprietary information.
Do not distribute, reproduce, or share without written permission.

Run Screen Menu (Errors and E-stop)



Seat / Top Position Error

- Indicates the press did not reach the expected proximity sensor
- This is a general (catch-all) fault for missed position
- Check for:
 - Mechanical binding
 - Misalignment
 - Gearbox or drivetrain issues
- Correct the issue, then re-home and resume operation

Errors and E-Stop Recovery

Stopping the Machine

- Press the **E-Stop** button at any time to immediately stop all machine motion.

Resetting the E-Stop

1. Twist the **E-Stop** clockwise to release it.
2. Wait 1–2 seconds for the system to reinitialize.
3. An error message will be displayed.

Error Types

Soft Errors

- Soft errors do not require press movement to recover.
- Correct the issue and cycle the E-Stop to continue.
- Examples include bullet orientation, reservoir sensors, and other external monitoring sensors.

Hard Errors

- Hard errors require operator intervention and press movement to recover.
- Use **Semi-Automated Mode** to clear the issue and return the press to a safe position.
- Examples include clutch faults, swage faults, or other conditions that stop the press mid-cycle.

Returning to Operation

- Correct the condition that caused the error.
- Clear the error message.
- Return the press to a safe operating position.
- Resume operation when ready.

Terminate Run

- Press **Terminate Run** to immediately end the current run.
- The machine will return to the Main Menu.
- The press will not move when terminating a run.

Notes

- Always correct the issue before restarting the machine.
- The press may stop at any point in the cycle.
- Error messages will identify the condition that caused the stop.

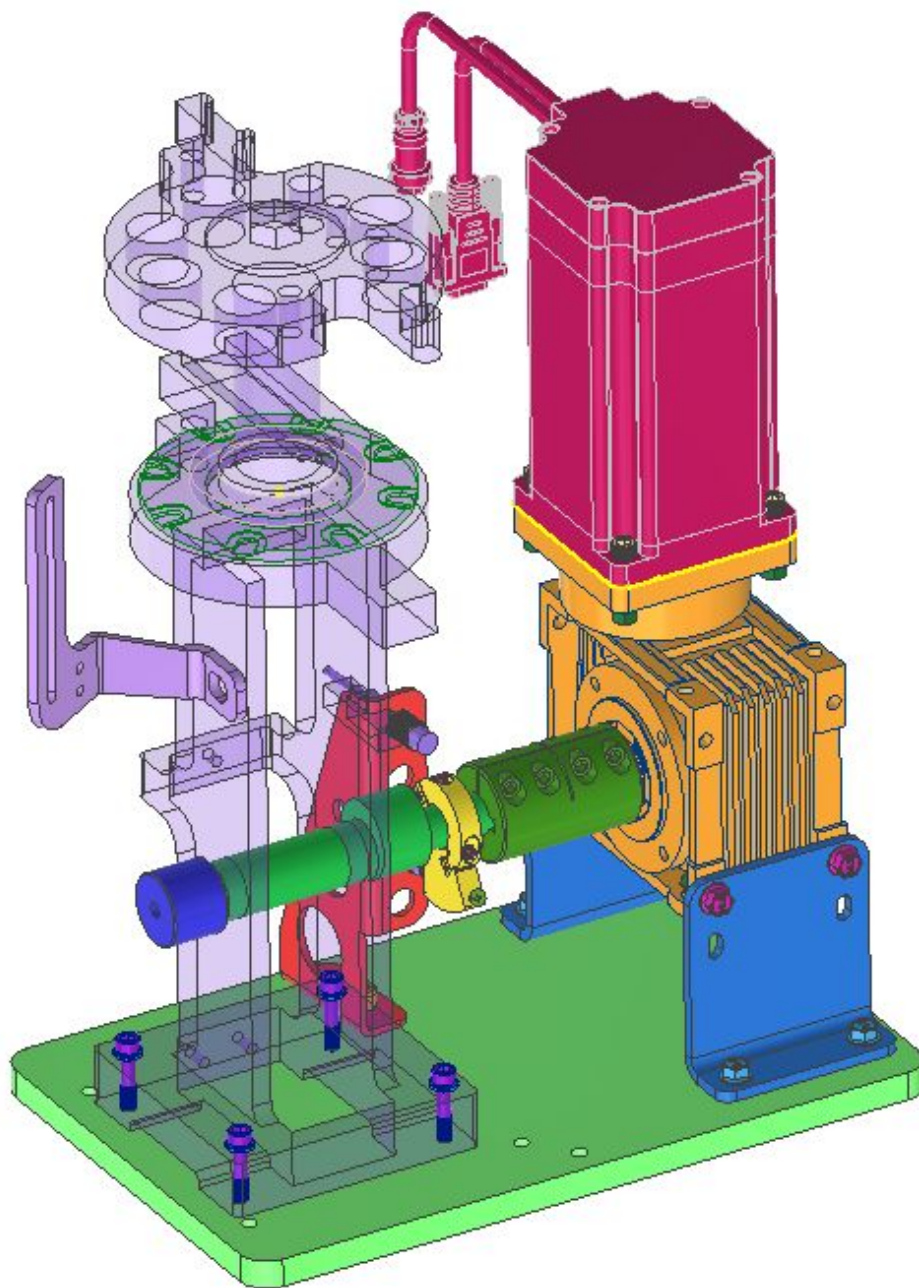
Quick Start Complete

At this point, your system is assembled and ready for setup.

Please refer to the **main manual** for operation, tuning, and advanced features of the AutoDrive and related systems.

- Navigate to the **Load Development** screen
- Set up and adjust your dies (X, Y, Z)
- Continue setup as needed for your application

You do not need to follow every section of the manual—use only what applies to your setup.



Confidential – Proprietary Information

© Gross Engineering LLC. All rights reserved.

This document contains confidential and proprietary information.
Do not distribute, reproduce, or share without written permission.